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TALEN ENERGY

AUG 0 3 2016

U. S. Nuclear Regulatory Commission
Attn: Document Control Desk
Washington, DC 20555-0001

10 CFR 50.73

SUSQUEHANNA STEAM ELECTRIC STATION
LICENSEE EVENT REPORT 50-387/2016-019-00
UNIT 1 LICENSE NO. NPF-14
PLA-7505

Docket No. 50-387

Attached is Licensee Event Report (LER) 50-387/2016-019-00. The LER reports a condition concerning Reactor Coolant Pressure Boundary leakage. This condition was determined to be reportable in accordance with 10 CFR 50.73(a)(2)(ii)(A), 10 CFR 50.73(a)(2)(i)(A), and 10 CFR 50.73(a)(2)(i)(B), as a condition resulting in a principal safety barrier degradation, plant shutdown required by Technical Specifications, and as a condition prohibited by Technical Specifications.

There were no actual consequences to the health and safety of the public as a result of this event.

This letter contains no new regulatory commitments.

for J. A. Franke
J. A. Franke

Attachment: LER 50-387/2016-019-00

Copy: NRC Region I
Mr. J. E. Greives, NRC Sr. Resident Inspector
Ms. T. E. Hood, NRC Project Manager
Mr. M. Shields, PA DEP/BRP

NRC FORM 366 (11-2015) U.S. NUCLEAR REGULATORY COMMISSION

LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

APPROVED BY OMB: NO. 3150-0104 EXPIRES: 10/31/2018

1. FACILITY NAME	2. DOCKET NUMBER	3. PAGE
Susquehanna Steam Electric Station Unit 1	05000387	1 of 4

4. TITLE
Pressure Boundary Leakage from an Inadequate Weld Repair in Small Bore Pump Seal Vent Piping

5. EVENT DATE			6. LER NUMBER			7. REPORT DATE			8. OTHER FACILITIES INVOLVED	
MONTH	DAY	YEAR	YEAR	SEQUENTIAL NUMBER	REV NO.	MONTH	DAY	YEAR	FACILITY NAME	DOCKET NUMBER 05000
06	06	2016	2016	- 019	- 00	08	03	2016	FACILITY NAME	DOCKET NUMBER 05000

9. OPERATING MODE: 2

10. POWER LEVEL: 009

11. THIS REPORT IS SUBMITTED PURSUANT TO THE REQUIREMENTS OF 10 CFR §: *(Check all that apply)*

☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☒ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)
☐ 20.2203(a)(2)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
☐ 20.2203(a)(2)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(v)(A)	☐ 73.71(a)(4)
☐ 20.2203(a)(2)(iii)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(5)
☐ 20.2203(a)(2)(iv)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(1)
☐ 20.2203(a)(2)(v)	☒ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(i)
☐ 20.2203(a)(2)(vi)	☒ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(ii)
	☐ 50.73(a)(2)(i)(C)	☐ OTHER	Specify in Abstract below or in NRC Form 366A

12. LICENSEE CONTACT FOR THIS LER

LICENSEE CONTACT	TELEPHONE NUMER (Include Area Code)
Regina Genovese - Nuclear Regulatory Affairs	(570) 542-2980

13. COMPLETE ONE LINE FOR EACH COMPONENT FAILURE DESCRIBED IN THIS REPORT

CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX	CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX
B	AD	P	Flowserve	Y					

14. SUPPLEMENTAL REPORT EXPECTED		15. EXPECTED SUBMISSION DATE	MONTH	DAY	YEAR
☐ YES *(If yes, complete 15. EXPECTED SUBMISSION DATE)*	☒ NO				

ABSTRACT *(Limit to 1400 spaces, i.e., approximately 15 single-spaced typewritten lines)*

On June 6, 2016, at 0556 hours during a drywell entry, a leak was reported on the Unit 1 B Reactor Recirculation Pump Lower Seal Vent Line. The leak was identified at the inboard pipe-to-union weld and required replacement prior to returning to service. The affected piping weld is for 3/4-inch piping, Quality Class Q (ASME Class 2) and is part of the Reactor Coolant Pressure Boundary.

The cause of the event was determined to be the result of a crack in the pipe-to-union weld on the pump side of the connection. The crack was determined to have initiated and propagated as a result of an undetectable lack of fusion at the weld root along with an unrecognized cyclic or vibratory loading. A complete replacement of the pump seal cooler assembly to remove the welded connection was completed prior to the restart of Unit 1.

The condition was immediately reported as a plant shutdown required by Technical Specifications (TS) pursuant to 10 CFR 50.72(b)(2)(i) and degraded condition pursuant to 10 CFR 50.72(b)(3)(ii)(A) (EN 51983). This event is being reported pursuant to 10 CFR 50.73(a)(2)(ii)(A), 10 CFR 50.73(a)(2)(i)(A), and 10 CFR 50.73(a)(2)(i)(B), as a condition resulting in a principal safety barrier degradation, plant shutdown required by Technical Specifications, and as a condition prohibited by Technical Specifications.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
Susquehanna Steam Electric Station, Unit 1	05000387	2016	- 019	- 00

NARRATIVE

CONDITIONS PRIOR TO EVENT

Unit 1 – Mode 2, 9 percent Rated Thermal Power
Unit 2 – Mode 1, 100 percent Rated Thermal Power

There were no other structures, systems, or components that were inoperable at the start of the event that contributed to the event.

EVENT DESCRIPTION

Susquehanna Unit 1 commenced a manual shutdown on June 5, 2016 for a maintenance outage to investigate an increasing trend in unidentified drywell leakage. On June 6, 2016, at approximately 0556, with Unit 1 in Mode 2, leakage was identified from the inboard (pump-side) pipe-to-union weld on the Lower Seal Vent Line [EIIS Component Identifier: PSP] for Unit 1 Reactor Recirculation [EIIS System Identifier: AD] Pump [EIIS Component Identifier: P] 1B. The piping is classified as American Society of Mechanical Engineers (ASME) Class 2 and is part of the Reactor Coolant Pressure Boundary. The leak location is within the Reactor Recirculation loop isolation valves and is isolable from the reactor vessel [EIIS Component Identifier: RPV]. The reactor was in Mode 2 at the time of discovery.

A timeline of relevant events follows:

November 2015	Unit 1 Scram as a result of a Main Steam Isolation Valve (MSIV) [EIIS Component Identifier: ISV] closure. During drywell walk-downs, a leak was identified on the Reactor Recirculation Pump B Lower Seal Vent Line. The pipe-to-union connection was replaced.
March-April 2016	During Unit 1 Refueling and Inspection Outage (RIO), follow-up pipe support inspections were conducted and a welded shim was installed to prevent movement of Lower Seal Vent Line.
June 2016	Unit 1 was down-powered to investigate an increasing trend in unidentified drywell leakage. Actual unidentified drywell leakage prior to shutdown was approximately 0.53 GPM. The inboard (pump-side) pipe-to-union weld on Lower Seal Vent Line for the Reactor Recirculation Pump B was identified as leaking.

This event was reported pursuant to 10 CFR 50.72(b)(2)(i) and 10 CFR 50.72(b)(3)(ii)(A) (EN 51983). This Licensee Event Report (LER) is being communicated pursuant to 10 CFR 50.73(a)(2)(ii)(A), 10 CFR 50.73(a)(2)(i)(A), and 10 CFR 50.73(a)(2)(i)(B), as a condition resulting in a principal safety barrier degradation, plant shutdown required by Technical Specifications, and as a condition prohibited by Technical Specifications.

NRC FORM 366A (11-2015) U.S. NUCLEAR REGULATORY COMMISSION	APPROVED BY OMB: NO. 3150-0104 EXPIRES: 10/31/2018
LICENSEE EVENT REPORT (LER) CONTINUATION SHEET	Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Privacy and Information Collections Branch (T-5 F53), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by internet e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
Susquehanna Steam Electric Station, Unit 1	05000387	2016	- 019	- 00

NARRATIVE

CAUSE OF EVENT

The cause of the Reactor Coolant Pressure Boundary leakage at the Reactor Recirculation Pump B Lower Seal Vent Line was determined to be a result of a crack in the pipe-to-union weld on the pump side of the connection. Based on investigation and laboratory testing, the crack was determined to have initiated and propagated as a result of an undetectable lack of fusion at the weld root along with an unrecognized cyclic or vibratory loading. Additionally, during the previous repairs on the pipe-to-union weld location, cold-working grinding was performed, inducing a compressive stress which contributed to crack initiation.

ANALYSIS/SAFETY SIGNIFICANCE

Technical Specification Limiting Condition for Operation (LCO) 3.4.4, RCS Operational Leakage, requires zero pressure boundary leakage and less than or equal to 5 gallons per minute (GPM) unidentified drywell leakage when operating in Modes 1, 2 and 3. At the time of discovery of the Reactor Coolant Pressure Boundary leak on the Lower Seal Vent Line, Unit 1 was in Mode 2. As such, the associated condition for Technical Specification (TS) 3.4.4 was not met, requiring operator actions to initiate a plant shutdown.

The 5 GPM Technical Specification limit for unidentified leakage is a small fraction of the calculated flow from a critical crack in the primary system piping. Crack behavior from experimental programs show that leakage rates of hundreds of gallons per minute will precede crack instability. As such, the Technical Specification limit allows time for corrective action to be taken before the Reactor Coolant Pressure Boundary could be significantly compromised. Prior to commencing the down-power to investigate the increasing trend in unidentified drywell leakage, the actual leakage was approximately 0.53 GPM. There was no evidence available to plant operators at the time to substantiate that leakage was from the Reactor Coolant Pressure Boundary. Based on the visual inspection and the size of the crack on the Lower Seal Vent Line, the leakage from this location was likely the primary contributor to the unidentified drywell leakage. Given the actual unidentified drywell leakage prior to shutdown, relative to the TS limit of 5 GPM, there was no actual consequence to public health or safety.

CORRECTIVE ACTIONS

Completed corrective actions include replacement of the pump seal cooler assembly, eliminating the union weld connection and preventing reoccurrence of this type of failure by use of a one-piece forged pipe-to–union fitting. Additionally, the Lower Seal Vent Line configuration was modified to be more flexible in order to alleviate pipe stresses from vibration. Lastly, a procedure change will be completed to provide guidance on preventing potential cold-working through grinding on small-bore piping.

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Susquehanna Steam Electric Station, Unit 1	05000387	2016	- 019	- 00

COMPONENT INFORMATION

Pump Manufacturer: FLOWSERVE (Formerly BW/IP)
Vent Line: ¾-inch pipe to 1-inch socket connection
Vent Line Material: ASME Class II SA312

PREVIOUS SIMILAR EVENTS

LER 50-387/2015-009-00, PLA-7419, “Pressure Boundary Leakage from an Inadequate Weld Repair in Small Bore Pump Seal Vent Piping.” January 11, 2016.

LER 50-388/2015-004-00, PLA-7337, “Degraded Condition Due to Reactor Coolant Pressure Boundary Leakage Caused by Vibration and Stiff Pipe Connection.” June 10, 2015.

LER 50-387/2014-011-00, PLA-7286, “Degraded Condition Due to Reactor Coolant Pressure Boundary Leakage Caused by an Inadequate Weld.” February 11, 2015.